Trigeminal nerve

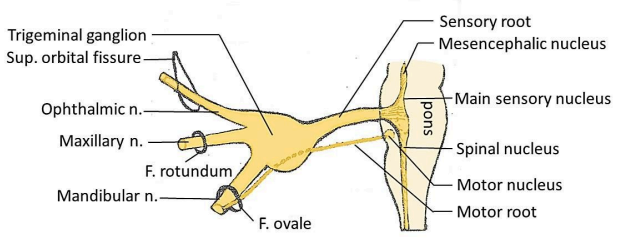
The great sensory nerve of the face, mouth, nose and sinuses: nuclei, the three divisions, their branches, ganglia and the referred pain it explains.

1. Nuclei and ganglion
2. Ophthalmic nerve (V1)
3. Maxillary nerve (V2)
4. Mandibular nerve (V3)
5. Ganglia attached to the trigeminal
6. Referred pain and clinical tests
7. On imaging

01 Nuclei and ganglion

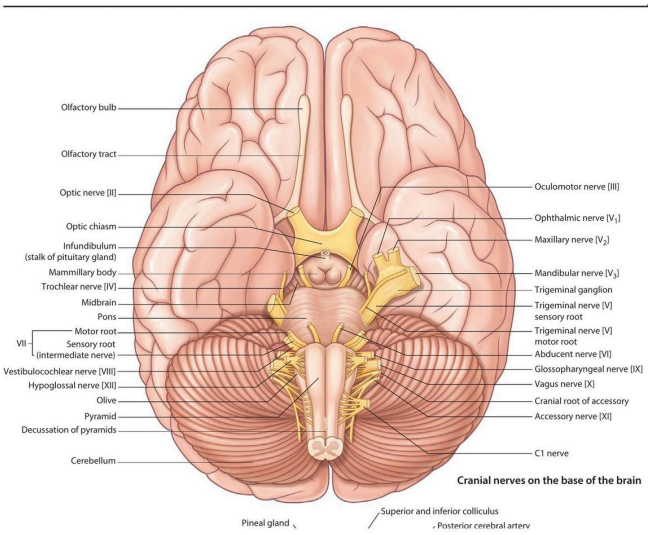
- Largest cranial nerve. Leaves the **lateral pons** as a large sensory root and a small motor root.
- **Nuclei:** mesencephalic (proprioception from muscles of mastication), **main (principal) sensory** (touch), **spinal** (pain and temperature, down to C2), **motor** (pons).
- **Trigeminal (Gasserian) ganglion** sits in **Meckel's cave** over the petrous apex (trigeminal impression).
- Divides into **V1 (ophthalmic)**, **V2 (maxillary)** and **V3 (mandibular)**; only V3 carries motor fibres.

> **CLINICAL PEARL** The spinal nucleus reaches C2, which explains why pain from the upper neck can be felt in the face and why Wallenberg syndrome causes facial pain-and-temperature loss.



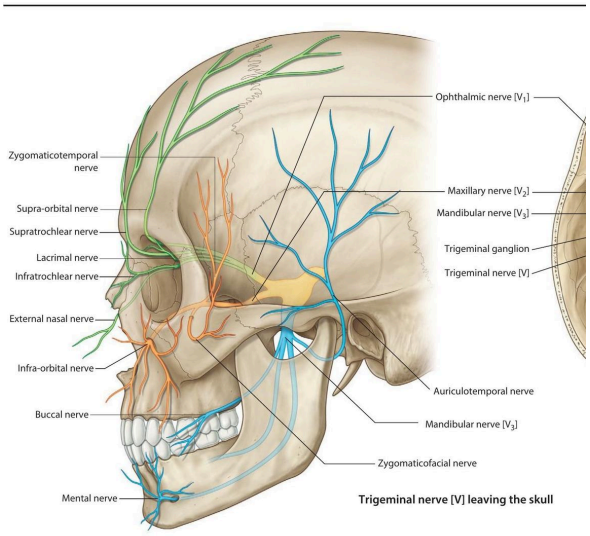
Trigeminal ganglion, divisions and brainstem nuclei

Dr. Sameh Doss handout, p. 148



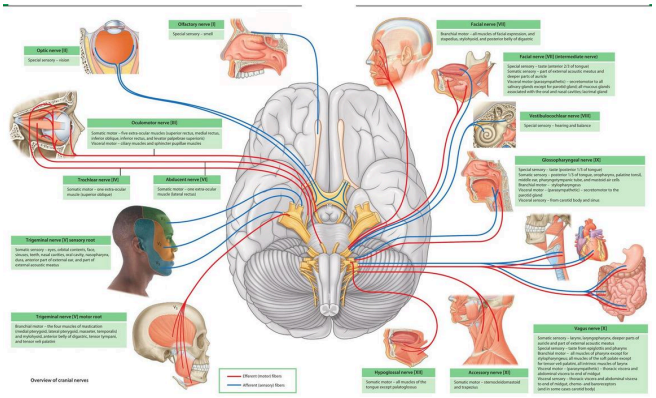
Cranial nerves on the base of the brain

Gray's Atlas of Anatomy, 2nd ed., p. 499



Trigeminal nerve [V] leaving the skull

Gray's Atlas of Anatomy, 2nd ed., p. 503



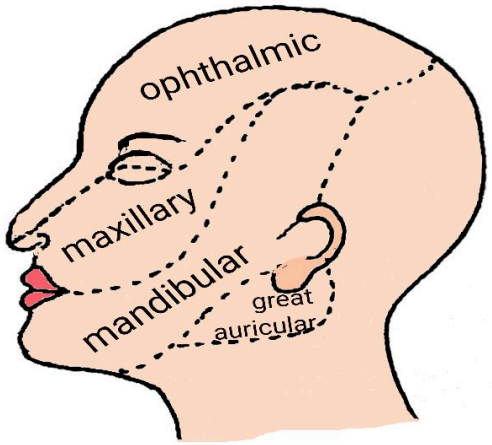
Overview of the cranial nerves

Gray's Atlas of Anatomy, 2nd ed., p. 584

02 Ophthalmic nerve (V1)

- Runs in the lateral wall of the cavernous sinus and enters the orbit through the **superior orbital fissure**.
- **Lacrimal** (lacrimal gland, lateral upper lid), **frontal** → supraorbital and supratrochlear (forehead, scalp, frontal sinus), **nasociliary** → anterior and posterior ethmoidal, long ciliary (cornea), infratrochlear.
- **Anterior ethmoidal nerve** ends as the **external nasal nerve**, supplying the skin of the nasal tip.

> **CLINICAL PEARL** Hutchinson's sign: herpes zoster vesicles on the nasal tip mean the nasociliary nerve is involved, so the eye is at risk.



Sensory territories of V1, V2 and V3 on the face

Dr. Sameh Doss handout, p. 47

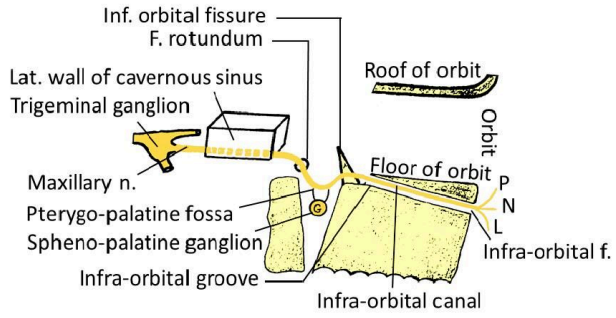
03 Maxillary nerve (V2)

- Purely sensory. Lateral wall of the cavernous sinus → **foramen rotundum** → **pterygopalatine fossa** → inferior orbital fissure → **infraorbital nerve** in the orbital floor → infraorbital foramen.
- **Branches:** meningeal; ganglionic (to the pterygopalatine ganglion); **zygomatic** (zygomaticofacial, zygomaticotemporal; carries lacrimal secretomotor fibres to the lacrimal nerve); **posterior, middle and**

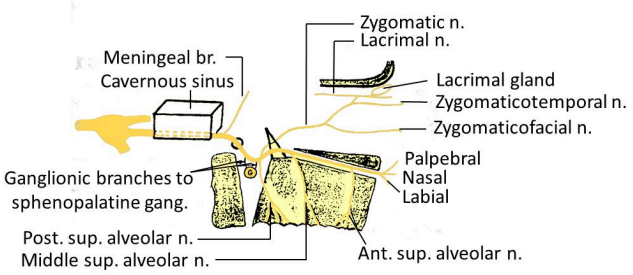
anterior superior alveolar (upper teeth, maxillary sinus); infraorbital (lower lid, cheek, side of nose, upper lip).

- **Through the pterygopalatine ganglion (sensory, no relay):** greater and lesser palatine (palate, tonsil), nasopalatine/long sphenopalatine (septum, front of palate), short sphenopalatine (lateral nasal wall), pharyngeal (nasopharynx, ET), orbital branches.

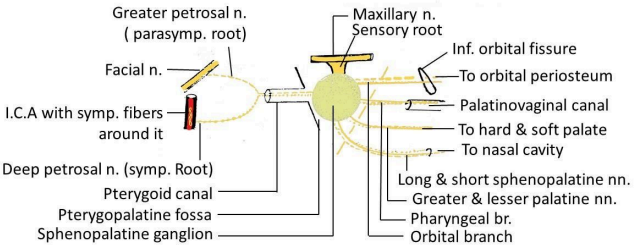
> **CLINICAL PEARL** Numbness of the cheek after a zygomatic or orbital floor fracture, or with a maxillary sinus tumour, means the infraorbital nerve is involved.



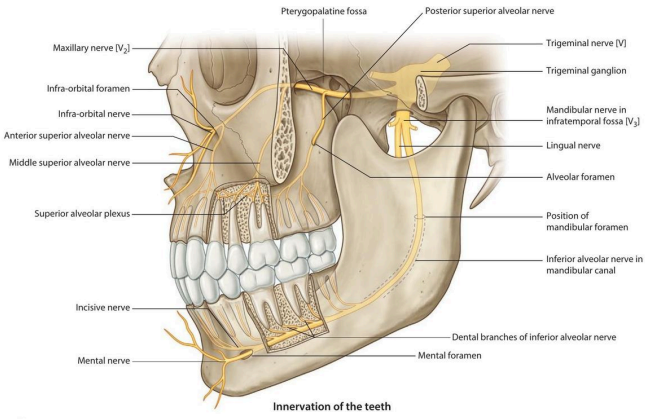
Maxillary nerve from the cavernous sinus to the face
Dr. Sameh Doss handout, p. 61



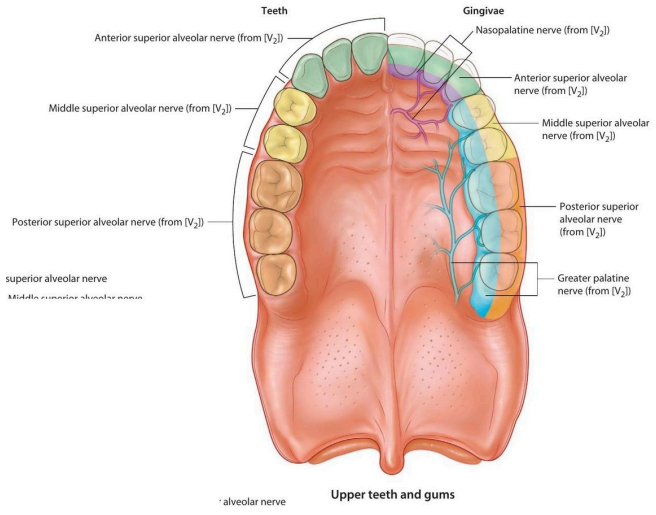
Branches of the maxillary nerve
Dr. Sameh Doss handout, p. 62



Pterygopalatine (sphenopalatine) ganglion
Dr. Sameh Doss handout, p. 63



576
Innervation of the teeth: superior and inferior alveolar nerves
Gray's Atlas of Anatomy, 2nd ed., p. 576

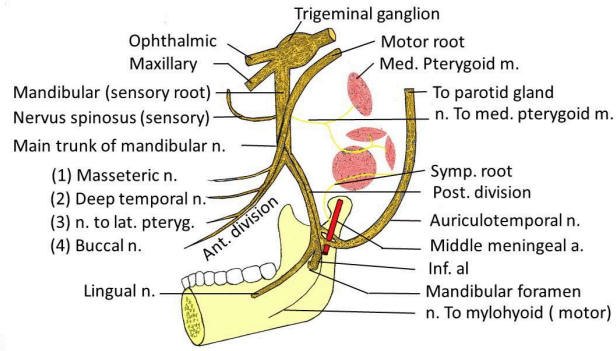


Upper teeth and gums: innervation
Gray's Atlas of Anatomy, 2nd ed., p. 577

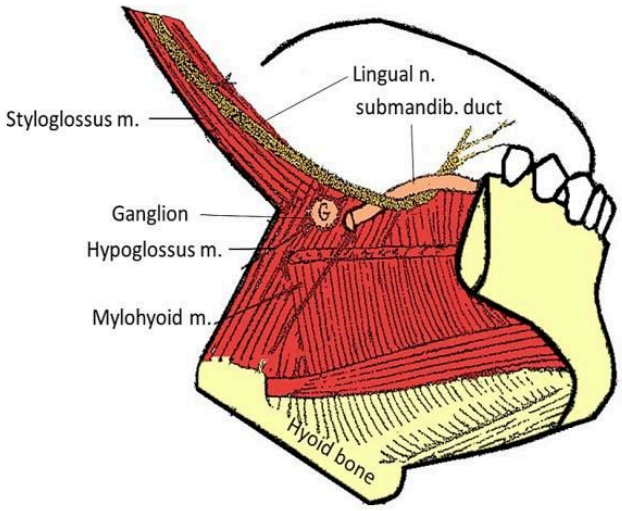
04 Mandibular nerve (V3)

- Mixed. Leaves through the **foramen ovale** into the infratemporal fossa.
- **Motor:** muscles of mastication, mylohyoid, anterior belly of digastric, tensor tympani, tensor veli palatini.
- **Sensory:** lower face, chin, lower lip (mental), lower teeth (inferior alveolar), anterior two-thirds of tongue and floor of mouth (lingual), cheek mucosa (buccal), temple, auricle, EAC, TM and TMJ (auriculotemporal).
- **Lingual nerve** lies just under the mucosa against the inner surface of the **third molar** socket; it then crosses the submandibular duct (lateral → below → medial).

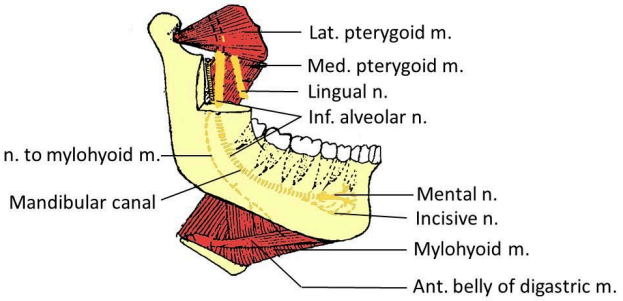
CLINICAL PEARL Third molar extraction can injure the lingual nerve; an inferior alveolar block numbs the lower lip and chin on that side.



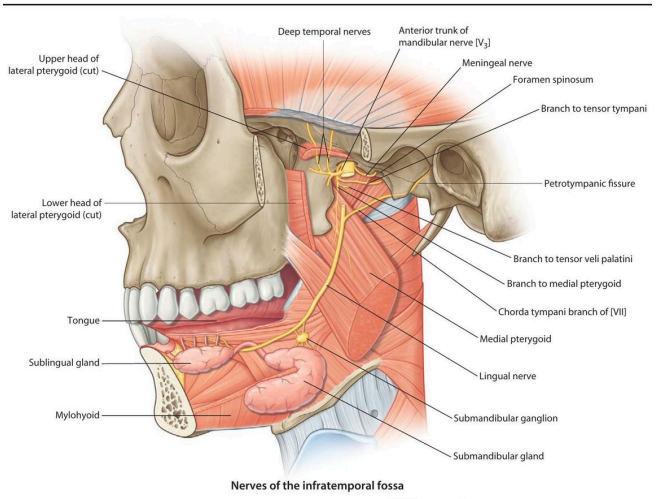
Mandibular nerve: trunk, divisions and branches
Dr. Sameh Doss handout, p. 55



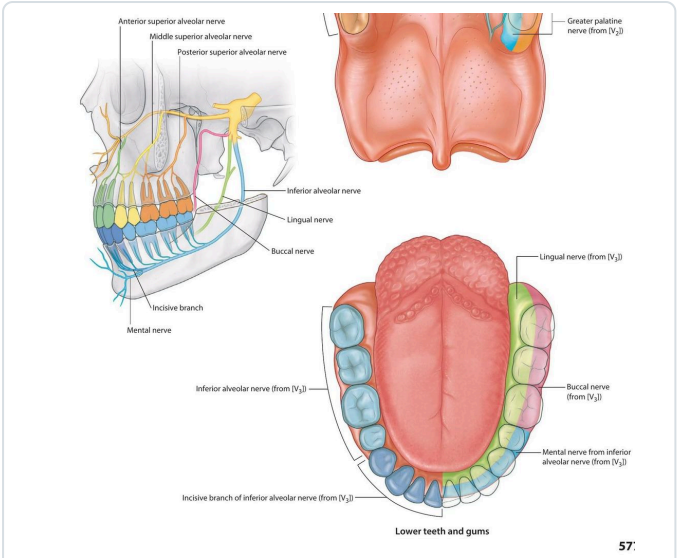
Lingual nerve and the submandibular ganglion
Dr. Sameh Doss handout, p. 57



Inferior alveolar nerve and nerve to mylohyoid
Dr. Sameh Doss handout, p. 58



Nerves of the infratemporal fossa: lingual, chorda tympani, otic ganglion
Gray's Atlas of Anatomy, 2nd ed., p. 532

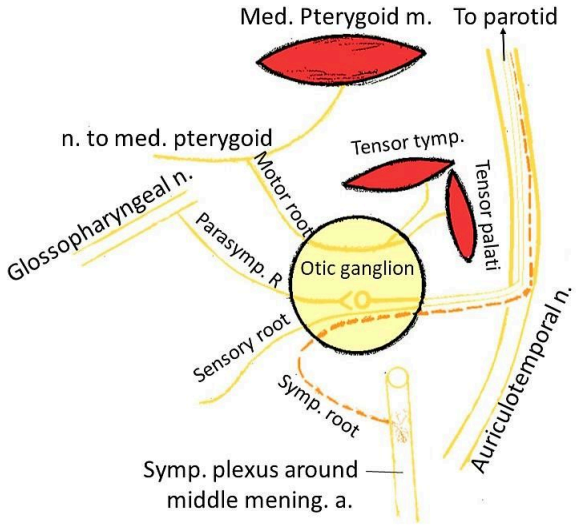


Lower teeth and gums: innervation

Gray's Atlas of Anatomy, 2nd ed., p. 577

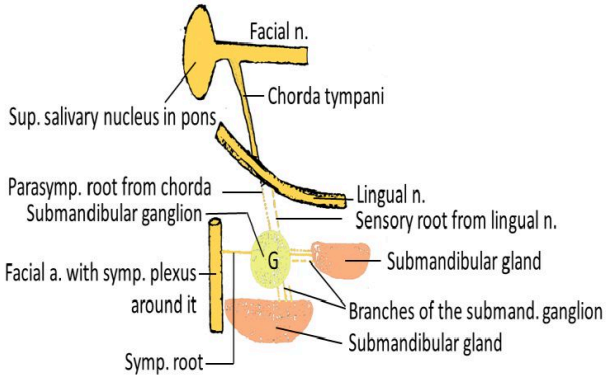
05 Ganglia attached to the trigeminal

- Four parasympathetic ganglia hang from the trigeminal divisions but get their secretomotor fibres from other nerves.
- **Ciliary** (V1): oculomotor parasympathetic → sphincter pupillae and ciliary muscle.
- **Pterygopalatine** (V2): greater petrosal (VII) → lacrimal, nasal, palatal glands. Largest of the four.
- **Otic** (V3): lesser petrosal (IX) → parotid via the auriculotemporal nerve.
- **Submandibular** (V3, hangs from the lingual): chorda tympani (VII) → submandibular and sublingual glands.
- Sympathetic and sensory fibres pass through each ganglion without relay.



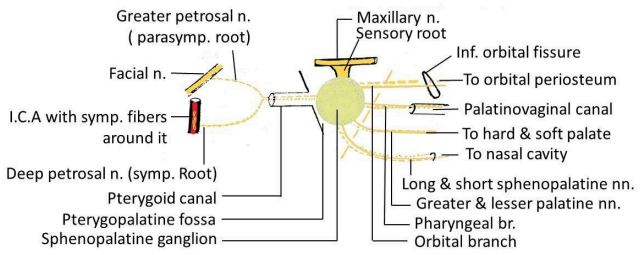
Otic ganglion and its connections

Dr. Sameh Doss handout, p. 59



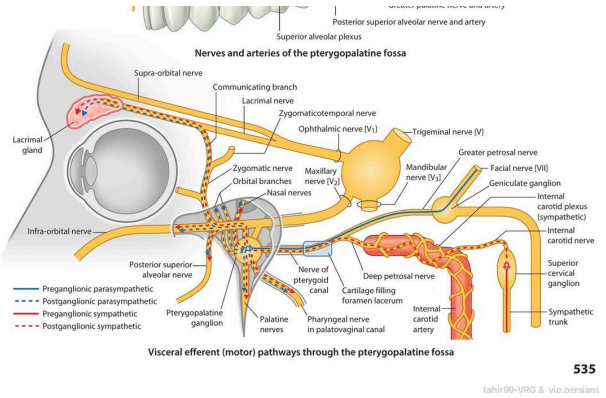
Submandibular ganglion and its connections

Dr. Sameh Doss handout, p. 60



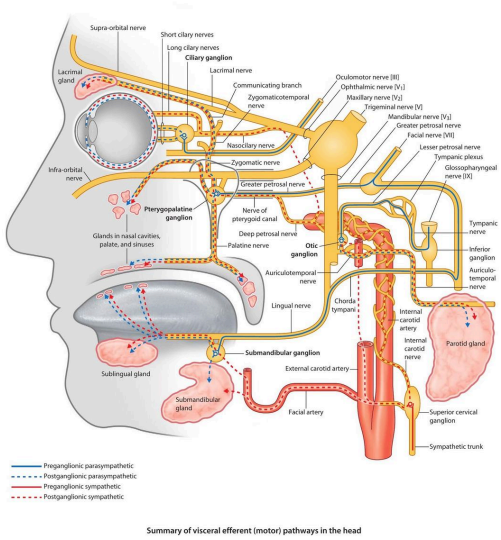
Pterygopalatine (sphenopalatine) ganglion

Dr. Sameh Doss handout, p. 63



Autonomic pathways through the pterygopalatine fossa

Gray's Atlas of Anatomy, 2nd ed., p. 535



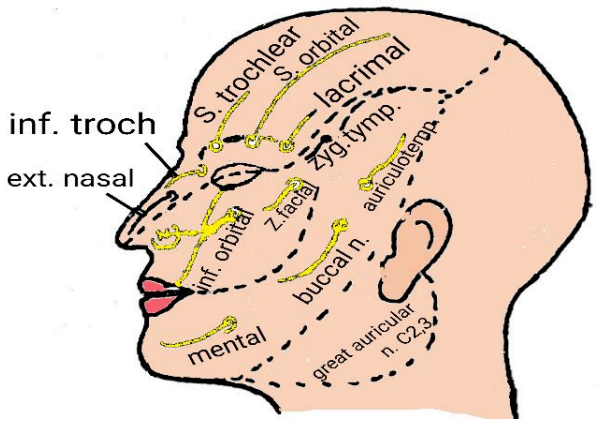
Summary of autonomic (visceral motor) pathways in the head

Gray's Atlas of Anatomy, 2nd ed., p. 586

06 Referred pain and clinical tests

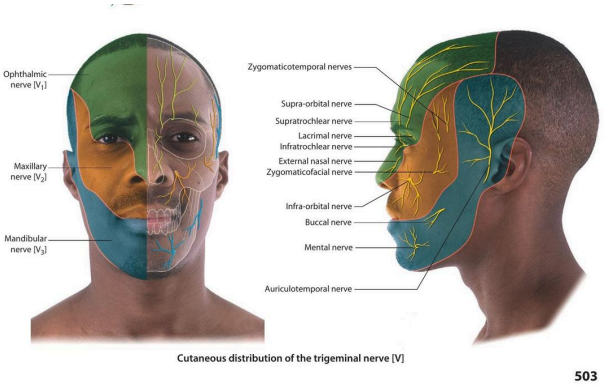
- **Referred otalgia via V3:** teeth (especially lower molars), TMJ, tongue, floor of mouth, retromolar trigone.
- **Referred pain via V2:** maxillary sinus and upper teeth to the cheek; sphenoid disease to the vertex.
- **Corneal reflex:** afferent V1 (nasociliary), efferent VII. Lost early with acoustic neuroma (V compression).
- **Jaw jerk:** afferent and efferent V3; brisk in upper motor neurone lesions.
- **Trigeminal neuralgia:** brief electric pains, usually V2 or V3, often from vascular compression at the root entry zone.

> **CLINICAL PEARL** Unexplained earache in an adult smoker: examine the oral cavity, oropharynx, hypopharynx and larynx before blaming the ear.



Cutaneous nerves of the face

Dr. Sameh Doss handout, p. 47



Cutaneous distribution of the trigeminal nerve [V]

Gray's Atlas of Anatomy, 2nd ed., p. 503

07 Radiological anatomy: On imaging

- **MRI with gadolinium** follows the nerve from the pons through Meckel's cave and the foramina: perineural tumour spread (adenoid cystic carcinoma, cutaneous SCC) enlarges and enhances the nerve and widens the foramen.
- **MRI CISS/FIESTA** shows vascular loops at the root entry zone in trigeminal neuralgia.

T Tables

Divisions of the trigeminal nerve

	V1 Ophthalmic	V2 Maxillary	V3 Mandibular
Exit	Superior orbital fissure	Foramen rotundum	Foramen ovale
Type	Sensory	Sensory	Mixed
Skin	Forehead, upper lid, dorsum and tip of nose	Lower lid, cheek, side of nose, upper lip	Lower lip, chin, skin over the body of the mandible (not the angle: C2–3), temple, part of auricle
Mucosa	Frontal, ethmoid, sphenoid sinuses; upper nasal cavity	Maxillary sinus, most of nasal cavity, palate, nasopharynx	Anterior 2/3 tongue, floor of mouth, cheek, lower gums
Ganglion attached	Ciliary	Pterygopalatine	Otic and submandibular
Reflex	Corneal (afferent)	Sneeze (afferent)	Jaw jerk (afferent and efferent)

Parasympathetic ganglia of the head

Ganglion	Preganglionic nerve	Hangs from	Supplies
Ciliary	Oculomotor (III)	Nasociliary (V1)	Sphincter pupillae, ciliary muscle
Pterygopalatine	Greater petrosal (VII) → vidian	Maxillary (V2)	Lacrimal, nasal, palatal and pharyngeal glands
Otic	Lesser petrosal (IX)	Mandibular (V3)	Parotid (via auriculotemporal)
Submandibular	Chorda tympani (VII) → lingual	Lingual (V3)	Submandibular and sublingual glands

Only the parasympathetic fibres synapse; sympathetic and sensory fibres pass through.

M Mnemonics

Foramina

Standing Room Only

Superior orbital fissure (V1), Rotundum (V2), Ovale (V3)

V3 anterior vs posterior division

Anterior Moves, Posterior Feels

Anterior division mainly motor (plus buccal); posterior mainly sensory (plus nerve to mylohyoid)

Parasympathetic ganglia

C-P-S-O = 3-7-7-9

Ciliary (III), Pterygopalatine (VII), Submandibular (VII), Otic (IX)

Q Rapid recall

1. **Four trigeminal nuclei?** Mesencephalic, main sensory, spinal and motor.
2. **Where is the trigeminal ganglion?** Meckel's cave over the petrous apex.
3. **Foramina of V1, V2 and V3?** Superior orbital fissure, foramen rotundum, foramen ovale.
4. **Which division of V carries motor fibres?** Only V3.
5. **What is Hutchinson's sign?** Zoster on the nasal tip: nasociliary nerve involved, eye at risk.
6. **Which nerve carries lacrimal secretomotor fibres to the lacrimal nerve?** Zygomaticotemporal branch of V2.
7. **Three superior alveolar nerves and their teeth?** Posterior (molars), middle (premolars), anterior (canine and incisors).
8. **Which ganglion is the largest parasympathetic ganglion?** Pterygopalatine (sphenopalatine), 4–5 mm.
9. **Only motor branch of the posterior division of V3?** Nerve to mylohyoid (mylohyoid and anterior belly of digastric).
10. **Afferent and efferent of the corneal reflex?** Afferent V1 (nasociliary), efferent VII.
11. **Course of the lingual nerve near the third molar?** Just under the mucosa on the inner side of the socket, at risk in extraction.
12. **Which muscles does V3 supply besides those of mastication?** Mylohyoid, anterior digastric, tensor tympani, tensor veli palatini.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.